Multi-Plan Telecommunications Cost Optimization – Complete Python Solution

1. Problem Understanding

A customer wants to select the most economical mobile recharge plan based on:

- Monthly data usage (GB).
- Monthly calling minutes.
- SMS requirements.
- Plan validity.
- Customer's budget.

The program evaluates multiple mobile plans, calculates excess usage charges, checks eligibility, and identifies the cheapest eligible plan.

Note: The mobile plans and prices below are imaginary examples for this assignment.

2. Computational Thinking

Input:

- Number of mobile plans.
- Customer's data usage.
- Calling minutes required.
- SMS required.
- Minimum validity.
- Maximum budget.
- Details of each mobile plan.

Processing:

1. Read the customer requirements.
2. Read the mobile plan details.
3. Calculate excess data, calling, and SMS charges.
4. Calculate the total cost of each plan.
5. Check validity and budget constraints.

6. Identify the cheapest eligible plan.

Output:

- Plan-wise cost details.
- Excess charges.
- Eligible and ineligible plans.
- Most economical eligible plan.

3. Assumptions and Rules

Each mobile plan contains:

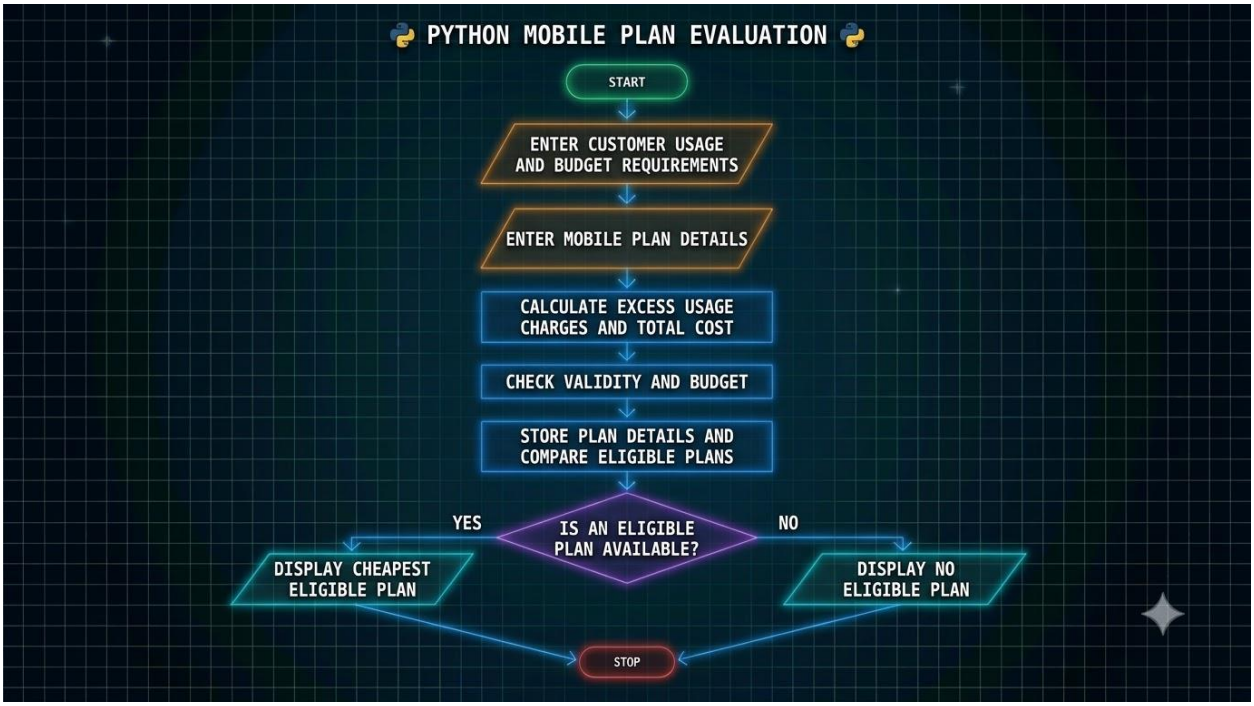
Parameter	Meaning
Price	Base recharge cost
Data	Included data in GB
Calls	Included calling minutes
SMS	Included SMS count
Validity	Number of days
Data excess rate	Cost per extra GB
Call excess rate	Cost per extra minute
SMS excess rate	Cost per extra SMS

A plan is eligible when:

- Its validity meets the customer's requirement.
- Its total cost does not exceed the customer's budget.

The cheapest eligible plan is selected.

4. Algorithm



5. Flowchart

START

Enter customer usage and budget requirements

Enter mobile plan details

Calculate excess usage charges and total cost

Check validity and budget

Store plan details and compare eligible plans

Is an eligible plan available?

YES

Display cheapest eligible plan

NO

Display no eligible plan

STOP

6. Pseudocode

START

INPUT customer data usage

INPUT customer calling minutes

INPUT customer SMS requirement

INPUT minimum validity

INPUT maximum budget

INPUT number of plans

CREATE empty plan list

FOR each plan:

    INPUT plan name

    INPUT base price

    INPUT included data

    INPUT included calls

    INPUT included SMS

    INPUT validity

    INPUT excess data rate

    INPUT excess call rate

    INPUT excess SMS rate

    excess_data = maximum of (usage - included data, 0)

    excess_calls = maximum of (calls required - included calls, 0)

```
    excess_sms = maximum of (SMS required - included SMS, 0)

    CALCULATE excess charges
    CALCULATE total cost

    IF validity >= minimum validity
       AND total cost <= budget:

        eligible = YES

    ELSE:

        eligible = NO

    STORE plan details

END FOR

DISPLAY all plans

FIND cheapest eligible plan

IF eligible plan exists:
    DISPLAY cheapest plan
ELSE:
    DISPLAY no eligible plan
```

STOP

7. Complete Python Program

```
# Multi-Plan Telecommunications Cost Optimization

# Customer requirements
usage = float(input("Enter required data usage in GB: "))
required_calls = int(input("Enter required calling minutes: "))
required_sms = int(input("Enter required SMS count: "))
minimum_validity = int(input("Enter minimum validity in days: "))
budget = float(input("Enter maximum budget: "))

number_of_plans = int(input("Enter number of mobile plans: "))

# List to store plan information
plans = []

# Input and process each plan
for i in range(number_of_plans):

    print("\nEnter details for Plan", i + 1)

    name = input("Enter plan name: ")
    price = float(input("Enter base price: "))
    data = float(input("Enter included data in GB: "))
    calls = int(input("Enter included calling minutes: "))
```

```
sms = int(input("Enter included SMS: "))
validity = int(input("Enter validity in days: "))

data_rate = float(input("Enter extra data charge per GB: "))
call_rate = float(input("Enter extra call charge per minute: "))
sms_rate = float(input("Enter extra SMS charge: "))

# Validate input
if (price < 0 or data < 0 or calls < 0 or
    sms < 0 or validity < 0 or data_rate < 0 or
    call_rate < 0 or sms_rate < 0):

    print("Invalid plan details. Plan skipped.")
    continue

# Calculate excess usage
extra_data = usage - data
if extra_data < 0:
    extra_data = 0

extra_calls = required_calls - calls
if extra_calls < 0:
    extra_calls = 0

extra_sms = required_sms - sms
if extra_sms < 0:
```

```
    extra_sms = 0

# Calculate excess charges
data_charge = extra_data * data_rate
call_charge = extra_calls * call_rate
sms_charge = extra_sms * sms_rate

total_extra = data_charge + call_charge + sms_charge

total_cost = price + total_extra

# Check eligibility
if validity >= minimum_validity and total_cost <= budget:
    eligible = "Yes"
else:
    eligible = "No"

# Store plan details in nested list
plan_details = [
    name, price, data, calls, sms, validity,
    extra_data, extra_calls, extra_sms,
    total_extra, total_cost, eligible
]

plans.append(plan_details)
```

```
# Display plan comparison
print("\n========== PLAN COMPARISON ==========")

for plan in plans:

    (name, price, data, calls, sms, validity,
     extra_data, extra_calls, extra_sms,
     total_extra, total_cost, eligible) = plan

    print("\nPlan Name:", name)
    print("Base Price: Rs.", price)
    print("Excess Data:", extra_data, "GB")
    print("Excess Calls:", extra_calls, "minutes")
    print("Excess SMS:", extra_sms)
    print("Total Extra Charges: Rs.", round(total_extra, 2))
    print("Total Cost: Rs.", round(total_cost, 2))
    print("Validity:", validity, "days")
    print("Eligible:", eligible)


# Find the cheapest eligible plan
cheapest_plan = ()
minimum_cost = budget + 1

for plan in plans:
```

```
    (name, price, data, calls, sms, validity,
     extra_data, extra_calls, extra_sms,
     total_extra, total_cost, eligible) = plan

    if eligible == "Yes":

        if total_cost < minimum_cost:
            minimum_cost = total_cost
            cheapest_plan = plan


# Display final result
print("\n========== FINAL RESULT ==========")

if len(cheapest_plan) > 0:

    (name, price, data, calls, sms, validity,
     extra_data, extra_calls, extra_sms,
     total_extra, total_cost, eligible) = cheapest_plan

    print("Most Economical Plan:", name)
    print("Final Cost: Rs.", round(total_cost, 2))
    print("Total Excess Charges: Rs.", round(total_extra, 2))
    print("Validity:", validity, "days")
```

else:

print("No eligible mobile plan found.")

print("Please increase your budget or reduce requirements.")

8. Sample Input

Customer requirements:

Enter required data usage in GB: 3

Enter required calling minutes: 500

Enter required SMS count: 100

Enter minimum validity in days: 28

Enter maximum budget: 300

Enter number of mobile plans: 3

Plan details:

Details	**Plan A**	**Plan B**	**Plan C**
Base price	₹199	₹249	₹179
Data (GB)	2	3	1.5
Calls (minutes)	500	600	300
SMS	100	100	50
Validity (days)	28	28	21
Extra data rate	₹20	₹20	₹20
Extra call rate	₹0.50	₹0.50	₹0.50
Extra SMS rate	₹1	₹1	₹1

9. Step-by-Step Calculations

Plan A

- Base price = ₹199

- Extra data = 3 − 2 = 1 GB
- Extra calls = 500 − 500 = 0
- Extra SMS = 100 − 100 = 0
- Extra charges = 1 × 20 = ₹20
- Total cost = 199 + 20 = ₹219
- Validity = 28 days
- Eligible = Yes

Plan B

- Base price = ₹249
- Extra data = 3 − 3 = 0 GB
- Extra calls = 0
- Extra SMS = 0
- Extra charges = ₹0
- Total cost = ₹249
- Validity = 28 days
- Eligible = Yes

Plan C

- Base price = ₹179
- Extra data = 3 − 1.5 = 1.5 GB
- Extra calls = 500 − 300 = 200 minutes
- Extra SMS = 100 − 50 = 50
- Data charge = 1.5 × 20 = ₹30
- Calling charge = 200 × 0.50 = ₹100
- SMS charge = 50 × 1 = ₹50
- Total cost = 179 + 30 + 100 + 50 = ₹359
- Validity = 21 days

- Eligible = No (validity is insufficient and cost exceeds budget).

10. Sample Output

========== PLAN COMPARISON ==========

Plan Name: Plan A

Base Price: Rs. 199.0

Excess Data: 1.0 GB

Excess Calls: 0 minutes

Excess SMS: 0

Total Extra Charges: Rs. 20.0

Total Cost: Rs. 219.0

Validity: 28 days

Eligible: Yes

Plan Name: Plan B

Base Price: Rs. 249.0

Excess Data: 0 GB

Excess Calls: 0 minutes

Excess SMS: 0

Total Extra Charges: Rs. 0.0

Total Cost: Rs. 249.0

Validity: 28 days

Eligible: Yes

Plan Name: Plan C

Base Price: Rs. 179.0

Excess Data: 1.5 GB

Excess Calls: 200 minutes

Excess SMS: 50

Total Extra Charges: Rs. 180.0

Total Cost: Rs. 359.0

Validity: 21 days

Eligible: No

========== FINAL RESULT ==========

Most Economical Plan: Plan A

Final Cost: Rs. 219.0

Total Excess Charges: Rs. 20.0

Validity: 28 days

Final answer: Plan A

Plan A is the most economical eligible plan because it costs ₹219, compared with Plan B at ₹249.

11. Test Cases

Test case	Condition	Expected result
1	Usage exactly matches plan	No excess charge
2	Usage exceeds data limit	Extra data charge calculated
3	Calling exceeds plan limit	Extra calling charge calculated
4	Validity below requirement	Plan ineligible
5	Cost exceeds budget	Plan ineligible
6	No plan satisfies requirements	No eligible plan found

12. Assigned Syllabus Concepts Used

Syllabus topic	Application
Computational thinking	Breaking plan selection into smaller steps
Algorithms	Step-by-step plan evaluation
Flowcharts	Visual representation of program logic
Pseudocode	Writing program logic in simple language
Variables	usage, budget, total_cost
Data types	Integer, float, string
Operators	Arithmetic and comparison operators
Conditional statements	if, else
Loops	for loop
Strings	Plan names and eligibility messages
Pattern generation	Not required for this problem
Lists	Storing mobile plans
Nested lists	Storing multiple plan details
Tuples	Storing and unpacking plan details
Tuple operations	Accessing and unpacking stored plan values

Important correction for a fully validated version: The program skips invalid plan entries, but the customer requirements and number of plans are assumed to be valid. For a classroom assignment, you can mention this assumption.